

FPGA-Based Smart Traffic Controller System

Project Overview

The FPGA-Based Smart Traffic Controller System is designed to improve traffic management at a two-road intersection by using a Finite State Machine (FSM) implemented in Verilog HDL. Unlike conventional traffic signal systems that operate with fixed timing, the proposed controller considers vehicle presence on the side road through a sensor input. Based on this input, the controller dynamically changes the traffic light sequence while maintaining safe traffic operation.

Objectives

Design and implement a smart traffic controller using Verilog HDL on FPGA, verify functionality through simulation, perform synthesis and timing analysis, and demonstrate efficient hardware implementation.

Design Methodology

Requirements analysis, FSM design, Verilog implementation, simulation with a testbench, synthesis using Quartus Prime Lite, and timing verification using TimeQuest Timing Analyzer.

System Architecture

Modules include Clock, Reset, Vehicle Sensor, FSM, Timer/Counter, Main Road Lights, and Side Road Lights.

Functional Description

The controller cycles through Main Green, Main Yellow, Side Green, and Side Yellow states based on sensor input and timer values, ensuring safe traffic flow.

Simulation and Verification

Simulation validates reset behavior, state transitions, sensor operation, and output traffic signals using a Verilog testbench and waveform analysis.

Synthesis and Timing Analysis

The design is synthesized in Intel Quartus Prime Lite. Resource utilization and timing reports confirm successful implementation with timing constraints satisfied.

Applications

Smart cities, road intersections, intelligent transportation systems, industrial traffic control, and parking management.

Advantages

High-speed execution, reconfigurability, deterministic operation, low latency, scalability, and efficient FPGA resource utilization.

Future Enhancements

Emergency vehicle priority, adaptive timing, camera-based detection, IoT connectivity, and AI-based traffic prediction.

Software Tools

Intel Quartus Prime Lite, ModelSim Intel FPGA Edition, Verilog HDL, GitHub.

Conclusion

The project demonstrates a reliable FPGA-based smart traffic controller using an FSM architecture. Functional simulation, synthesis, and timing analysis verify that the design is suitable for practical FPGA implementation.